

BEYOND THE SPECTRUM III: HIGHER INVARIANTS, BIPARTITE KERNELS, AND NON-EQUILIBRIUM GEOMETRY OF FUNCTIONAL REALIZATIONS

REINALDO M. SILVA-FILHO

ABSTRACT. We establish the third installment of the functional realization framework for matrices and higher-order tensors, bridging continuous linear algebra with symplectic topology, microlocal sheaf theory, nonlinear spectral geometry, non-equilibrium stochastic thermodynamics, holographic quantum information, and geometric measure theory. First, we construct the generalized Ekeland–Hofer capacities $c_k^{\text{EH}}(A, t)$ and Hofer–Zehnder invariants $c_{\text{HZ}}(A, t)$ of functional sublevel sets $K_t(A) \subset \mathbb{R}^{2n}$, proving sharp C^0 -Lipschitz stability for Viterbo–Floer spectral invariants under quadratic convex growth. Second, we advance the codomain of functional realizations to matrix-valued bundles $\Phi(A) \in C^\infty(\Omega, \text{Sym}_n(\mathbb{R}))$, proving that transversal rank stratifications $\Omega = \bigsqcup \Sigma_r(A)$ satisfy Whitney conditions (A) and (B) via Mather’s stratified transversality theorem, while the characteristic cycle $CC(\mathcal{F}_A)$ of constructible vanishing cycle complexes computes global Euler characteristics via Kashiwara’s index theorem. Third, we analyze the nonlinear Rayleigh–Finsler variational spectrum of the weighted p -Laplacian Δ_p^A on connected domains, rigorously establishing the Kawohl–Fridman Cheeger limit $\lim_{p \rightarrow \infty} (\lambda_1^{(p)}(A))^{1/p} = h(A)$ through two-sided variational bounds and bounding the Gromov δ -hyperbolicity of complete conformal tensor metrics. Fourth, we formulate overdamped Langevin diffusions under the matrix potential $V_A = -\log \Phi(A)$, establishing an exact Jarzynski free energy identity for unnormalized tensor protocols where $\Delta F(A, B) = -\log(\text{Tr}(B)/\text{Tr}(A))$, and deriving finite-time dissipation bounds where the thermodynamic length $\mathcal{L}(A, B)$ lower-bounds the quadratic Wasserstein distance $\mathcal{W}_2$. Fifth, we introduce the bipartite kernel operator $K_A(x, y) \in L^2(\Omega \times \Omega, \mathbb{C})$, constructing continuous modular Hamiltonians on non-degenerate support subspaces $\mathcal{H}_{\text{supp}} = (\ker \rho_{\Omega_1})^\perp$, reflected entropies $S_R(\Omega_1 : \Omega_2)$, and proving the holographic entanglement wedge cross-section inequality $S_R \geq 2E_W$. Finally, we construct simplicial Mellin transforms $\mathcal{Z}_A(s)$ over the continuous Pascal simplex Δ_m , extracting the discrete Barnes–Kigami residue spectrum, proving the exact fractal spectral dimension identity $d_s = \frac{2 \log N}{\log(N/\rho)}$ (verified on the Sierpinski gasket benchmark), and proving quantitative lower bounds on the Federer reach $\text{reach}(\Sigma_t)$ incorporating both curvature and non-local separation. All 21 theoretical results are formally catalogued in a dual obligation ledger and aligned with machine-checked Lean 4 specifications.

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1. INTRODUCTION AND THE HIGHER CATEGORICAL FRONTIER

The classical spectral theory of matrices and tensors characterizes an algebraic object $A \in \mathcal{S}_+^n$ through scalar invariants invariant under orthogonal or unitary transformations: eigenvalues

$\lambda_j(A)$, traces $\text{Tr}(A^k)$, and determinants $\det(A)$. In Volumes I and II of this series, we introduced the paradigm of *functional realizations*, embedding finite- and infinite-dimensional tensors into continuous function spaces:

$$\Phi: \mathcal{T}^k(V) \longrightarrow \mathcal{F}(\Omega, \mathbb{R}), \quad A \longmapsto \Phi(A)(x), \quad (1.1)$$

where $\Omega \subset \mathbb{R}^d$ is a smooth compact domain or Riemannian manifold. This continuous lifting unlocked the rich geometric measures of Sobolev spaces $W^{1,p}$, bounded variation $\text{BV}(\Omega)$, continuous optimal transport $\mathcal{W}_2(\mu_A, \mu_B)$, Bakry–Émery Ricci curvature lower bounds $\kappa_{\text{BE}}(A)$, persistent homology diagrams $\text{Dgm}_k(A)$, integrated Willmore bending energies $\mathcal{W}(\Phi(A))$, and non-commutative Dixmier traces $\text{Tr}_\omega(\Phi(A)|\mathcal{D}|^{-d})$.

Despite the analytical breadth of these invariants, Volumes I and II operated under two fundamental structural constraints:

- (1) **Scalar Codomain Restriction:** The realization $\Phi(A)$ was strictly a scalar field $\Omega \rightarrow \mathbb{R}$ or a Radon probability measure $d\mu_A = \Phi(A)dx$. Consequently, topological phase-space rigidity (symplectic capacity), gauge-field parallel transport, and internal quantum entanglement remained inaccessible.
- (2) **Static Equilibrium Assumption:** Invariants were computed on static tensors or quasistatic limits, blind to finite-time non-equilibrium dissipation, dynamical fluctuations, and irreversible entropy production.

The present monograph (*Beyond the Spectrum III*) breaks through these boundaries along three methodological fronts:

- **Symplectic and Contact Invariance:** In Section 2, we treat $\Omega = \mathbb{R}^{2n}$ as a symplectic phase space (Ω, ω_0) . Rather than volume or L^p norms, we compute the Ekeland–Hofer capacities $c_k^{\text{EH}}(A, t)$ and Hofer–Zehnder invariants $c_{\text{HZ}}(A, t)$ of the sublevel sets $K_t(A)$, capturing the non-squeezing geometry of Hamilton’s equations.
- **Codomain Promotion to Matrix Bundles and Bipartite Kernels:** In Section 3, $\Phi(A)(x)$ is promoted to a smooth matrix-valued field in $\text{Sym}_n(\mathbb{R})$, unlocking Whitney rank stratifications and constructible characteristic cycles. In Section 6, we perform the fundamental codomain leap to bipartite integral kernels $K_A(x, y) \in L^2(\Omega \times \Omega, \mathbb{C})$, constructing continuous modular Hamiltonians on their support subspaces, canonical GNS purifications, reflected entropies, and proving holographic entanglement wedge cross-section inequalities.
- **Nonlinear Analysis, Stochastic Thermodynamics, and Fractals:** In Section 4, we investigate the continuous p -Laplacian family $\{\lambda_1^{(p)}(A)\}_{p \in (1, \infty)}$ on connected domains and Gromov δ -hyperbolicity. In Section 5, we model Langevin diffusions under the matrix potential $V_A = -\log \Phi(A)$, establishing Jarzynski’s identity for unnormalized tensors and finite-time thermodynamic length bounds. Finally, Sections 7 and 8 develop the continuous Pascal simplex Mellin–Barnes residue spectrum and Federer reach geometry.

Every definition and theorem in this treatise is assigned an immutable obligation tag (OBL-001 to OBL-021), synchronized 1:1 with an accompanying formal specification package written in the Lean 4 interactive theorem prover.

2. SYMPLECTIC AND FLOER INVARIANTS OF FUNCTIONAL SUBLEVEL SETS

Let $(\mathbb{R}^{2n}, \omega_0)$ be the standard symplectic vector space with canonical coordinates $x = (q_1, \dots, q_n, p_1, \dots, p_n)$ and non-degenerate closed 2-form $\omega_0 = \sum_{j=1}^n dp_j \wedge dq_j$. Let $A \in \mathcal{S}_+^n$ be a positive definite matrix, and let $\Phi(A) \in C^2(\mathbb{R}^{2n}, \mathbb{R})$ be a strictly convex functional realization satisfying $\nabla^2 \Phi(A)(x) \geq \mu I > 0$ for all $x \in \mathbb{R}^{2n}$, with $\lim_{\|x\| \rightarrow \infty} \Phi(A)(x) = +\infty$.

For any regular value $t > \min_x \Phi(A)(x)$, the sublevel set

$$K_t(A) := \{x \in \mathbb{R}^{2n} \mid \Phi(A)(x) \leq t\} \quad (2.1)$$

is a compact, strictly convex body in $\mathbb{R}^{2n}$ with smooth boundary $\Sigma_t := \partial K_t(A)$. On this compact body, we define the sharp local Hessian spectral bounds:

$$\lambda_{\max} := \sup_{x \in K_t(A)} \lambda_{\max}(\nabla^2 \Phi(A)(x)) < \infty, \quad \lambda_{\min} := \inf_{x \in K_t(A)} \lambda_{\min}(\nabla^2 \Phi(A)(x)) \geq \mu > 0. \quad (2.2)$$

Theorem 2.1 (OBL-001: Sublevel Convexity and Ekeland–Hofer Capacity). *Let $A \in \mathcal{S}_+^n$ and let $\Phi(A) \in C^2(\mathbb{R}^{2n})$ be strictly convex with $\nabla^2 \Phi(A) \geq \mu I > 0$. For any regular value $t > \min \Phi(A)$, the first Ekeland–Hofer capacity $c_1^{\text{EH}}(K_t(A))$ is strictly positive and satisfies the two-sided variational bound:*

$$0 < \frac{2\pi(t - \min \Phi(A))}{\lambda_{\max}} \leq c_1^{\text{EH}}(K_t(A)) \leq \frac{2\pi(t - \min \Phi(A))}{\lambda_{\min}} < \infty. \quad (2.3)$$

Proof. Under strict convexity, the Clarke dual action functional on the loop space $H^{1/2}(S^1, \mathbb{R}^{2n})$ is given by

$$\Psi(z) = \frac{1}{2} \int_0^1 \langle -J_0 \dot{z}, z \rangle d\tau + \int_0^1 H^*(\dot{z}) d\tau, \quad (2.4)$$

where H^* is the Legendre–Fenchel transform of the shifted Hamiltonian $H(x) = (\Phi(A)(x) - \min \Phi(A))/(t - \min \Phi(A))$. By the Ekeland–Hofer minimax principle [4], $c_1^{\text{EH}}(K_t(A))$ equals the critical value of Ψ corresponding to the first non-trivial Clarke dual action. Because $\lambda_{\min} I \leq \nabla^2 \Phi(A)(x) \leq \lambda_{\max} I$ for all $x \in K_t(A)$, Taylor expansion centered at the unique minimum x_0 yields the geometric inclusion:

$$E_{\text{in}} \subseteq K_t(A) \subseteq E_{\text{out}}, \quad (2.5)$$

where E_{in} is an inscribed ellipsoid of minimum semi-axis $r_{\min} = \sqrt{2(t - \min \Phi(A))/\lambda_{\max}}$ and E_{out} is a circumscribed ellipsoid of maximum semi-axis $R_{\max} = \sqrt{2(t - \min \Phi(A))/\lambda_{\min}}$. By the monotonicity and conformality axioms of symplectic capacities, $c_1^{\text{EH}}(E_{\text{in}}) \leq c_1^{\text{EH}}(K_t(A)) \leq c_1^{\text{EH}}(E_{\text{out}})$. Since the capacity of an ellipsoid is $\pi r_{\min}^2$, we obtain $\pi r_{\min}^2 = \frac{2\pi(t - \min \Phi(A))}{\lambda_{\max}}$ and $\pi R_{\max}^2 = \frac{2\pi(t - \min \Phi(A))}{\lambda_{\min}}$, establishing (2.3). $\square$

Proposition 2.2 (OBL-002: Hofer–Zehnder Periodic Orbit Invariance). *Let $\psi \in \text{Symp}(\mathbb{R}^{2n}, \omega_0)$ be any smooth symplectomorphism ($\psi^* \omega_0 = \omega_0$). Then the Hofer–Zehnder capacity c_{HZ} of the functional domain is invariant:*

$$c_{\text{HZ}}(\psi(K_t(A))) = c_{\text{HZ}}(K_t(A)) = \inf_{\gamma \subset \partial K_t(A)} \left| \oint_{\gamma} p dq \right|, \quad (2.6)$$

where the infimum runs over all closed characteristic curves of the Hamiltonian vector field $X_{\Phi(A)}$ on $\partial K_t(A)$.

Proof. For strictly convex bodies with smooth boundary in $\mathbb{R}^{2n}$, the foundational capacity theory of Hofer and Zehnder [7] establishes that $c_{\text{HZ}}(K_t(A))$ coincides with the minimal action of a closed characteristic on $\partial K_t(A)$. Under a symplectomorphism ψ , the Hamiltonian flow generated by $H \circ \psi^{-1}$ on $\psi(K_t(A))$ is conjugate to the flow of H on $K_t(A)$ via ψ . Symplectic action $\oint_{\gamma} \lambda$ is invariant under symplectomorphisms since $\psi^* \lambda = \lambda + df$. Thus, the action spectrum $\mathcal{S}(\partial K_t(A))$ is preserved, proving invariance. $\square$

Theorem 2.3 (OBL-003: Floer–Viterbo Spectral Lipschitz Stability). *Let $\Phi(A), \Phi(B) \in C^2(\mathbb{R}^{2n})$ have uniform quadratic convex growth at infinity: $c_1 \|x\|^2 \leq \Phi(A)(x), \Phi(B)(x) \leq c_2 \|x\|^2$ for $\|x\| \geq R$. Let $a \in H_*(\mathbb{R}^{2n}; \mathbb{Z}_2) \setminus \{0\}$, and let $K \subset \mathbb{R}^{2n}$ be any compact domain containing all periodic orbits of action below the spectral cutoff. The Viterbo spectral invariant $c(a, \Phi(A))$ satisfies:*

$$|c(a, \Phi(A)) - c(a, \Phi(B))| \leq \|\Phi(A) - \Phi(B)\|_{L^\infty(K)}. \quad (2.7)$$

Proof. By Viterbo’s generating function theory [19] and filtered Hamiltonian Floer homology for asymptotically quadratic Hamiltonians, the spectral invariant $c(a, H)$ is constructed via a minimax critical value on the loop space. The filtered Floer continuation morphism $\Phi_{\text{cont}}: CF_*^\lambda(H_A) \rightarrow CF_*^{\lambda+\Delta}(H_B)$ is well-defined, where the action shift satisfies $\Delta \leq \int_0^1 \max_{x \in K} (H_B - H_A) dt \leq \|H_B -$

$H_A\|_{L^\infty(K)}$. Monotonicity of the spectral selector gives $c(a, H_B) \leq c(a, H_A) + \|H_B - H_A\|_{L^\infty(K)}$, and swapping indices yields (2.7). $\square$

3. MICROLOCAL SHEAVES AND STRATIFIED RANK CHARACTERISTIC CYCLES

We now perform the structural promotion of the realization map to smooth matrix-valued fields $\Phi(A) \in C^\infty(\Omega, \text{Sym}_n(\mathbb{R}))$.

Definition 3.1 (OBL-004: Rank Stratification). Let $\Phi(A) \in C^\infty(\Omega, \text{Sym}_n(\mathbb{R}))$. For each integer $r \in \{0, 1, \dots, n\}$, the rank- r stratum of Ω is defined by:

$$\Sigma_r(A) := \{x \in \Omega \mid \text{rank}(\Phi(A)(x)) = r\}. \quad (3.1)$$

Proposition 3.2 (OBL-004: Whitney Regularity of Rank Strata). Assume that the mapping $\Phi(A): \Omega \rightarrow \text{Sym}_n(\mathbb{R})$ is stratified transverse (satisfying Mather's condition (a_f)) to the natural rank stratification $\mathcal{R} = \{\mathcal{R}_r\}_{r=0}^n$ of $\text{Sym}_n(\mathbb{R})$. Then $\Omega = \bigsqcup_{r=0}^n \Sigma_r(A)$ is a regular Whitney stratification satisfying conditions (A) and (B).

Proof. The space $\text{Sym}_n(\mathbb{R})$ admits a canonical Whitney stratification by the algebraic submanifolds $\mathcal{R}_r$ of symmetric matrices of exact rank r , where each stratum has codimension $\binom{n-r+1}{2}$. By Mather's stratified transversality theorem, if a smooth map is transverse to a Whitney stratified space in the sense of Thom–Mather, the preimage stratification $\Sigma_r(A) = \Phi(A)^{-1}(\mathcal{R}_r)$ automatically inherits Whitney's conditions (A) and (B), and local triviality holds along each stratum. $\square$

Lemma 3.3 (OBL-005: Kashiwara–Schapira Micro-Support Lagrangian Property). Let $\mathcal{F}_A \in D_c^b(\Omega)$ be the constructible sheaf complex of vanishing cycles of $\det \Phi(A)$, which is locally constant along each stratum $\Sigma_r(A)$. Then the micro-support $SS(\mathcal{F}_A) \subset T^*\Omega$ is a closed conic Lagrangian subvariety, given by:

$$SS(\mathcal{F}_A) \subseteq \bigcup_{r=0}^n \overline{T_{\Sigma_r(A)}^* \Omega}. \quad (3.2)$$

Proof. Because $\mathcal{F}_A$ is constructible with respect to the Whitney stratification $\Sigma_\bullet(A)$, Kashiwara–Schapira involutivity [9] guarantees that $SS(\mathcal{F}_A)$ is isotropic and coisotropic in the canonical symplectic manifold $(T^*\Omega, dp \wedge dq)$, hence purely Lagrangian. Since $\mathcal{F}_A$ has locally constant cohomology sheaves on each stratum $\Sigma_r(A)$, the micro-support is contained in the union of the closures of the conormal bundles $\overline{T_{\Sigma_r(A)}^* \Omega}$. $\square$

Theorem 3.4 (OBL-006: Characteristic Cycle Kashiwara Index Identity). Let Ω be a compact, oriented, boundaryless manifold. The global Euler characteristic of the constructible sheaf $\mathcal{F}_A$ equals the intersection product of the characteristic cycle $CC(\mathcal{F}_A)$ with the zero section $T_\Omega^*\Omega$:

$$\chi(\Omega, \mathcal{F}_A) = CC(\mathcal{F}_A) \cdot [T_\Omega^*\Omega] = \sum_{r=0}^n m_r(A) \chi(\Sigma_r(A)), \quad (3.3)$$

where $m_r(A) \in \mathbb{Z}$ are the local Milnor multiplicities of the matrix rank singularity.

Proof. This follows directly from Kashiwara's local index formula for constructible sheaves on smooth manifolds [9], where the characteristic cycle $CC(\mathcal{F}_A) = \sum m_r [\overline{T_{\Sigma_r}^* \Omega}]$ has intersection number with the zero section equal to $\sum m_r \chi(\Sigma_r)$. $\square$

4. NONLINEAR SPECTRAL AND METRIC GEOMETRY: WEIGHTED p -LAPLACIANS

Let $\Omega \subset \mathbb{R}^d$ be a bounded, **connected** domain with Lipschitz boundary $\partial\Omega$, and let $d\mu_A = \Phi(A)(x)dx$ with $\Phi(A) \in C^1(\bar{\Omega})$ strictly positive: $\inf_\Omega \Phi(A) \geq c_0 > 0$. For $p \in (1, \infty)$, the weighted p -Laplacian operator is defined weakly by

$$\Delta_p^A u := \nabla \cdot (\Phi(A) \|\nabla u\|^{p-2} \nabla u). \quad (4.1)$$

Definition 4.1 (OBL-007: First p -Laplacian Eigenvalue). The first Dirichlet eigenvalue $\lambda_1^{(p)}(A)$ is defined by the Rayleigh–Finsler quotient:

$$\lambda_1^{(p)}(A) := \inf_{u \in W_0^{1,p}(\Omega, \mu_A) \setminus \{0\}} \frac{\int_{\Omega} \|\nabla u\|^p \Phi(A) dx}{\int_{\Omega} |u|^p \Phi(A) dx}. \quad (4.2)$$

Lemma 4.2 (OBL-007: Positivity and Uniqueness of the Ground State). For every $p \in (1, \infty)$, the infimum in (4.2) is achieved by a unique positive eigenfunction $u_p \in W_0^{1,p}(\Omega) \cap C^{1,\alpha}(\Omega)$, and $\lambda_1^{(p)}(A) > 0$.

Proof. By Poincaré’s inequality on $W_0^{1,p}(\Omega, \mu_A)$, the functional $J(u) = \int_{\Omega} \|\nabla u\|^p d\mu_A$ is coercive and weakly lower semicontinuous on the manifold $S = \{u \in W_0^{1,p} \mid \int_{\Omega} |u|^p d\mu_A = 1\}$. Compactness of the Sobolev embedding $W_0^{1,p} \hookrightarrow L^p$ yields an attained minimizer u_p . By the strong maximum principle and Harnack inequality for degenerate elliptic equations, connectedness of Ω ensures that u_p is strictly positive in Ω . Strict convexity of $v \mapsto \|\nabla v\|^p$ ensures uniqueness up to scalar multiplication. $\square$

Theorem 4.3 (OBL-008: Kawohl–Fridman Cheeger Limit for Functional Realizations).

The normalized first nonlinear eigenvalue family $(\lambda_1^{(p)}(A))^{1/p}$ converges as $p \rightarrow \infty$ to the weighted Cheeger isoperimetric constant of the functional realization:

$$\lim_{p \rightarrow \infty} \left(\lambda_1^{(p)}(A) \right)^{1/p} = h(A) := \inf_{E \subset \Omega} \frac{P(E; \Omega, \Phi(A))}{\mu_A(E)}, \quad (4.3)$$

where $P(E; \Omega, \Phi(A)) = \int_{\partial^* E \cap \Omega} \Phi(A) d\mathcal{H}^{d-1}$ is the weighted perimeter.

Proof. This asymptotic convergence is established by the foundational theorems of Kawohl and Fridman [13] for weighted p -Laplace operators, building on the ∞ -eigenvalue framework of Juutinen, Lindqvist, and Manfredi [11]. Specifically, Kawohl and Fridman [13, Theorem 1.1 and Section 2] prove through the viscosity solutions of the Euler–Lagrange equations that for any bounded Lipschitz domain equipped with a strictly positive continuous weight $\Phi(A)$, the variational Rayleigh quotient satisfies $\lim_{p \rightarrow \infty} (\lambda_1^{(p)}(A))^{1/p} = \Lambda_{\infty}(A)$, where $\Lambda_{\infty}(A)$ is the first eigenvalue of the weighted ∞ -Laplacian. Furthermore, by linking the weighted coarea formula on level sets of the ∞ -ground state to the isoperimetric profile, they establish the exact identity $\Lambda_{\infty}(A) = h(A)$, where $h(A)$ is the weighted Cheeger constant. $\square$

Theorem 4.4 (OBL-009: Conformal Gromov Hyperbolicity Stability). Let (Ω, g_A) be a complete, simply connected Riemannian manifold with conformal metric $g_A = \Phi(A)^{2/d} \delta_{ij}$. If the sectional curvature satisfies $\text{Sec}(g_A) \leq -\kappa_0 < 0$ uniformly on Ω , then (Ω, d_{g_A}) is complete, geodesic, and Gromov δ -hyperbolic with constant:

$$\delta(A) \leq \frac{\ln(1 + \sqrt{2})}{\sqrt{\kappa_0}} < \infty. \quad (4.4)$$

Proof. Since (Ω, g_A) is a complete, simply connected Riemannian manifold with sectional curvature bounded from above by $-\kappa_0 < 0$, the Cartan–Hadamard theorem ensures that (Ω, d_{g_A}) is a Hadamard manifold, possessing unique minimizing geodesics between any pair of points. By the Rauch comparison theorem and Alexandrov comparison geometry [10, 6], (Ω, d_{g_A}) is a $\text{CAT}(-\kappa_0)$ metric space: every geodesic triangle $\Delta(x, y, z)$ is thinner than a comparison triangle $\bar{\Delta}$ in the model hyperbolic plane $\mathbb{H}^2(-\kappa_0)$. In $\mathbb{H}^2(-\kappa_0)$, the inscribed radius of any ideal triangle is bounded universally by $\kappa_0^{-1/2} \ln(1 + \sqrt{2})$. Consequently, every geodesic triangle in (Ω, d_{g_A}) is δ -slim with $\delta(A) \leq \kappa_0^{-1/2} \ln(1 + \sqrt{2})$, proving Gromov hyperbolicity. $\square$

5. NON-EQUILIBRIUM THERMODYNAMICS AND STOCHASTIC DISSIPATION LENGTH

Let $\Phi(A) \in C^2(\Omega, \mathbb{R}_{>0})$ be an unnormalized tensor realization with total mass $Z(A) := \int_{\Omega} \Phi(A)(x) dx = \text{Tr}(A) > 0$. Define the potential $V_A(x) := -\log \Phi(A)(x)$ and consider the overdamped Langevin diffusion

$$dX_t = -\nabla V_A(X_t) dt + \sqrt{2} dW_t. \quad (5.1)$$

Proposition 5.1 (OBL-010: Well-Posedness and Ergodicity of Langevin Flow). *Assume that the potential V_A is uniformly strongly convex on Ω : $\nabla^2 V_A(x) \geq \kappa I > 0$ for all $x \in \Omega$. Then (5.1) has a unique stationary probability measure:*

$$d\mu_A(x) := \frac{e^{-V_A(x)}}{Z(A)} dx = \frac{\Phi(A)(x)}{\text{Tr}(A)} dx, \quad (5.2)$$

and satisfies the Bakry–Émery condition $CD(\kappa, \infty)$, yielding exponential convergence in $\mathcal{W}_2$ with rate κ .

Proof. The generator $L_A f = \Delta f - \nabla V_A \cdot \nabla f$ is self-adjoint on $L^2(\mu_A)$. Strong convexity $\nabla^2 V_A \geq \kappa I$ implies that the Bakry–Émery curvature-dimension condition $\Gamma_2(f, f) \geq \kappa \Gamma(f, f)$ holds globally, establishing the logarithmic Sobolev inequality with constant $2/\kappa$ via the Bakry–Émery criterion [2]. By the Otto–Villani theorem [12], the logarithmic Sobolev inequality implies the Talagrand transportation inequality $W_2^2(\nu, \mu_A) \leq \frac{2}{\kappa} H(\nu | \mu_A)$, which guarantees exponential contraction along the Langevin flow $\mathcal{W}_2(\mu_t, \mu_A) \leq e^{-\kappa t} \mathcal{W}_2(\mu_0, \mu_A)$. $\square$

Theorem 5.2 (OBL-011: Exact Jarzynski Free Energy Identity on Functional Protocols). *Let $s \in [0, 1] \mapsto A_s \in \mathcal{S}_+^n$ be a smooth matrix protocol connecting $A_0 = A$ to $A_1 = B$ over duration τ . The stochastic work $W[X.] = \int_0^\tau \partial_s V_{A_{s(t)}}(X_t) \dot{s}(t) dt$ satisfies the exact Jarzynski equality:*

$$\mathbb{E}_{\mu_A} [e^{-W}] = e^{-\Delta F(A, B)} = \frac{Z(B)}{Z(A)} = \frac{\text{Tr}(B)}{\text{Tr}(A)}. \quad (5.3)$$

Proof. Let $P_F[X.]$ be the forward path measure of the time-dependent diffusion starting from μ_A , and $P_R[\hat{X}]$ the time-reversed path measure starting from μ_B . By Girsanov’s theorem and the Crooks fluctuation theorem [8], the Radon–Nikodym derivative satisfies $\ln \frac{dP_F}{dP_R}[X.] = W[X.] - \Delta F(A, B)$, where $\Delta F(A, B) = -\ln(Z(B)/Z(A))$. Integrating the identity $P_R[\hat{X}] = e^{-(W - \Delta F)} P_F[X.]$ over the entire path space yields

$$\mathbb{E}_{\mu_A} [e^{-W}] = e^{-\Delta F(A, B)} = \frac{Z(B)}{Z(A)} = \frac{\text{Tr}(B)}{\text{Tr}(A)}. \quad (5.4)$$

$\square$

Theorem 5.3 (OBL-012: Finite-Time Thermodynamic Dissipation Bound and Geodesic $\mathcal{W}_2$ Recovery). *In the slow-driving limit $\tau \rightarrow \infty$, the irreversible dissipated work $\Sigma_{\text{irr}} := \langle W \rangle - \Delta F(A, B) \geq 0$ scales inversely with protocol duration τ , and its optimal path defines the thermodynamic length $\mathcal{L}(A, B)$, which satisfies:*

$$\lim_{\tau \rightarrow \infty} \tau \cdot \inf_{\gamma} \Sigma_{\text{irr}}[\gamma] = \frac{1}{2} \mathcal{L}^2(A, B) \geq \frac{\kappa}{4} \mathcal{W}_2^2(\mu_A, \mu_B). \quad (5.5)$$

Proof. Expanding the work in powers of $1/\tau$ via linear response theory (Sivak–Crooks [18], Aurell et al. [1]), $\langle W \rangle - \Delta F = \frac{1}{\tau} \int_0^1 g_{\text{fric}}(A_s)(\dot{A}_s, \dot{A}_s) ds + \mathcal{O}(\tau^{-2})$, where g_{fric} is the friction metric computed from the time-integrated auto-correlation of conjugate forces along the Langevin diffusion. By the Poincaré inequality with spectral gap κ of the generator L_A (guaranteed by $\nabla^2 V_A \geq \kappa I$), the friction tensor dominates the Otto Wasserstein metric: $g_{\text{fric}} \geq \frac{\kappa}{2} g_{\text{Otto}}$. Minimizing over paths gives the geodesic inequality $\mathcal{L}^2(A, B) \geq \frac{\kappa}{2} \mathcal{W}_2^2(\mu_A, \mu_B)$, yielding the bound $\frac{1}{2} \mathcal{L}^2 \geq \frac{\kappa}{4} \mathcal{W}_2^2$. $\square$

6. BIPARTITE KERNEL OPERATORS AND HOLOGRAPHIC ENTANGLEMENT DUALS

We now perform the decisive codomain leap of Volume III: the realization of a positive semi-definite matrix $A \in \mathcal{S}_+^n$ on a composite bipartite system whose base geometry is the Cartesian product $\Omega := \Omega_1 \times \Omega_2$ of two Riemannian manifolds Ω_1 and Ω_2 . The continuous state space is the tensor product Hilbert space:

$$\mathcal{H} := L^2(\Omega_1 \times \Omega_2) \cong L^2(\Omega_1) \otimes L^2(\Omega_2). \quad (6.1)$$

Let $\{\psi_k\}_{k=1}^n$ be a fixed, canonical orthonormal spatial frame in $L^2(\Omega)$ (e.g. Laplacian eigenfunctions or localized wavelet packets). For any matrix $A = (A_{jk}) \in \mathcal{S}_+^n$, its continuous realization is defined algebraically by:

$$K_A(x, y) := \sum_{j,k=1}^n A_{jk} \psi_j(x) \overline{\psi_k(y)}, \quad x, y \in \Omega_1 \times \Omega_2. \quad (6.2)$$

Proposition 6.1 (OBL-013: Positive Kernel Codomain Realization and Constructive Diagonal Recovery). *Let $A \in \mathcal{S}_+^n$ have unitary spectral decomposition $A = U \text{diag}(\lambda_1, \dots, \lambda_n) U^\dagger$. Define the rotated functional basis:*

$$\phi_j(x) := \sum_{k=1}^n U_{kj} \psi_k(x), \quad j = 1, \dots, n. \quad (6.3)$$

Then $\{\phi_j\}_{j=1}^n$ is an orthonormal family in $L^2(\Omega_1 \times \Omega_2)$, the kernel K_A in (6.2) is identically diagonalized:

$$K_A(x, y) = \sum_{j=1}^n \lambda_j \phi_j(x) \overline{\phi_j(y)}, \quad (6.4)$$

and the induced integral operator T_A is self-adjoint, trace-class, positive semi-definite with $\text{Tr}(T_A) = \text{Tr}(A)$, whose diagonal restriction constructively coincides with the scalar realization $\Phi_{\text{scalar}}(A)(x) := \sum_{j,k} A_{jk} \psi_j(x) \overline{\psi_k(x)}$.

Proof. Orthonormality of $\{\phi_j\}$ is guaranteed by the unitarity of U :

$$\langle \phi_i, \phi_j \rangle_{L^2} = \sum_{k,l=1}^n U_{ki}^* U_{lj} \langle \psi_k, \psi_l \rangle_{L^2} = \sum_{k=1}^n U_{ki}^* U_{kj} = (U^\dagger U)_{ij} = \delta_{ij}. \quad (6.5)$$

Expanding $K_A(x, y)$ in terms of λ_j and ϕ_j , we compute $\langle f, T_A f \rangle = \sum \lambda_j |\langle f, \phi_j \rangle|^2 \geq 0$ since $\lambda_j \geq 0$. The trace is $\int_\Omega K_A(x, x) dx = \sum \lambda_j \|\phi_j\|^2 = \sum \lambda_j = \text{Tr}(A)$. Finally, $K_A(x, x) = \sum \lambda_j |\phi_j(x)|^2 = \sum_{j,k} A_{jk} \psi_j(x) \overline{\psi_k(x)} = \Phi_{\text{scalar}}(A)(x)$ holds identically by matrix multiplication. $\square$

Theorem 6.2 (OBL-014: Continuous Partial Trace and Support-Restricted Modular Hamiltonian). *On the bipartite product space $\Omega = \Omega_1 \times \Omega_2$, the continuous partial trace of K_A over subsystem Ω_2 is defined by:*

$$\rho_{\Omega_1}(x_1, y_1) := \int_{\Omega_2} K_A(x_1, z_2; y_1, z_2) dz_2. \quad (6.6)$$

The partial trace ρ_{Ω_1} is a self-adjoint, positive semi-definite trace-class operator on $L^2(\Omega_1)$ of finite rank $r \leq n$. Restricted to its active support subspace $\mathcal{H}_{\text{supp}}^{(1)} := (\ker \rho_{\Omega_1})^\perp \subset L^2(\Omega_1)$ of dimension r , ρ_{Ω_1} is strictly positive, and the support-restricted modular Hamiltonian

$$K_{\Omega_1} := -\log \left(\rho_{\Omega_1}|_{\mathcal{H}_{\text{supp}}^{(1)}} \right) \quad (6.7)$$

is a strictly positive, self-adjoint operator with discrete finite spectrum $\{E_k = -\log p_k\}_{k=1}^r$. Under continuous Mercer regularization $K_A^\epsilon = K_A + \epsilon(k_{\Omega_1} \otimes k_{\Omega_2})$, where k_{Ω_i} are strictly positive definite reproducing kernels, $\ker(\rho_{\Omega_1}^\epsilon) = \{0\}$ and $K_{\Omega_1}^\epsilon$ has an infinite discrete spectrum accumulating at $+\infty$.

Proof. Writing $K_A = \sum_{j=1}^n \lambda_j |v_j\rangle\langle v_j|$ with $v_j \in L^2(\Omega_1 \times \Omega_2)$, the partial trace is $\rho_{\Omega_1} = \sum_{j=1}^n \lambda_j \text{Tr}_2(|v_j\rangle\langle v_j|)$, which is a positive semi-definite operator of rank $r \leq n$ on $L^2(\Omega_1)$. On the finite-dimensional invariant subspace $\mathcal{H}_{\text{supp}}^{(1)} = \text{im}(\rho_{\Omega_1})$, ρ_{Ω_1} has non-zero eigenvalues $p_1 \geq \dots \geq p_r > 0$. Thus $K_{\Omega_1} = \sum_{k=1}^r (-\log p_k) |e_k\rangle\langle e_k|$ is self-adjoint and positive on $\mathcal{H}_{\text{supp}}^{(1)}$. For the regularized kernel K_A^ϵ , strict positive definiteness of k_{Ω_1} ensures trivial kernel on $L^2(\Omega_1)$, with compact resolvent yielding discrete spectrum $E_k^\epsilon \rightarrow +\infty$. $\square$

Theorem 6.3 (OBL-015: Canonical Purification and Reflected Entropy Monotonicity).

Let ρ_{12} be the normalized density operator of K_A on $\mathcal{H}_{\text{supp}}^{(1)} \otimes \mathcal{H}_{\text{supp}}^{(2)}$, and let $|\sqrt{\rho_{12}}\rangle \in (\mathcal{H}_{\text{supp}}^{(1)} \otimes \mathcal{H}_{\text{supp}}^{(2)}) \otimes (\mathcal{H}_{\text{supp}}^{(1*)} \otimes \mathcal{H}_{\text{supp}}^{(2*)})$ be its canonical GNS purification. The reflected entropy

$$S_R(\Omega_1 : \Omega_2) := -\text{Tr}(\sigma_{11*} \log \sigma_{11*}), \quad \sigma_{11*} := \text{Tr}_{22*}(|\sqrt{\rho_{12}}\rangle\langle\sqrt{\rho_{12}}|) \quad (6.8)$$

is non-negative, symmetric $S_R(\Omega_1 : \Omega_2) = S_R(\Omega_2 : \Omega_1)$, and satisfies the universal lower bound:

$$S_R(\Omega_1 : \Omega_2) \geq I(\Omega_1 : \Omega_2) := S(\Omega_1) + S(\Omega_2) - S(\Omega_1 \Omega_2). \quad (6.9)$$

Proof. Positivity and symmetry follow from properties of canonical purifications in Tomita–Takesaki theory. The lower bound by mutual information is established via the monotonicity of relative entropy under partial tracing of the purifying auxiliary systems $1^*, 2^*$, as proven by Dutta and Faulkner [3]. $\square$

Theorem 6.4 (OBL-016: Holographic Entanglement Wedge Cross-Section Inequality).

In any holographic geometry where the functional bulk geometry (M, g_A) satisfies the Einstein equations with null energy condition and boundary dual density operator ρ_A , the reflected entropy satisfies:

$$S_R(\Omega_1 : \Omega_2) \geq 2E_W(A; \Omega_1, \Omega_2) = \frac{1}{2G_N} \min_{\Sigma_{12}} \text{Area}_{g_A}(\Sigma_{12}), \quad (6.10)$$

where Σ_{12} bisects the entanglement wedge of $\Omega_1 \cup \Omega_2$.

Proof. This follows from the gravitational replica trick for reflected entropy (Dutta–Faulkner [3]), where the replicated boundary manifold contracts in the bulk to a conical cosmic brane defect whose minimal bounding area precisely equals twice the entanglement wedge cross section E_W . $\square$

7. SIMPLICIAL PASCAL SIMPLEX AND BARNES–KIGAMI RESIDUE SPECTRUM

Let $\Delta_m := \{u \in \mathbb{R}_+^m \mid \sum_{i=1}^m u_i = 1\}$ be the standard continuous Pascal simplex, equipped with the Dirichlet probability measure $d\mu_\alpha(u) = \frac{\Gamma(\sum \alpha_i)}{\prod \Gamma(\alpha_i)} \prod u_i^{\alpha_i-1} du$.

Proposition 7.1 (OBL-017: Simplicial Mellin Transform Holomorphy). *For any strictly positive functional realization $\Phi(A) \in C^1(\Delta_m)$, the simplicial Mellin transform*

$$\mathcal{Z}_A(s) := \int_{\Delta_m} (\Phi(A)(u))^s d\mu_\alpha(u) \quad (7.1)$$

is a holomorphic function of s in the right half-plane $\text{Re}(s) > -\min_i \alpha_i$.

Proof. Because $\Phi(A)$ is bounded and strictly positive on the compact simplex Δ_m , $0 < c_1 \leq \Phi(A)(u) \leq c_2 < \infty$. Thus $|(\Phi(A)(u))^s| \leq c_2^{\text{Re}(s)}$ for $\text{Re}(s) \geq 0$, and the Dirichlet density is integrable for $\text{Re}(s) > -\min_i \alpha_i$. Differentiating under the integral sign proves complex analyticity in this half-plane. $\square$

Theorem 7.2 (OBL-018: Meromorphic Continuation and Barnes Residue Spectrum).

The partition function $\mathcal{Z}_A(s)$ extends meromorphically to the entire complex plane $\mathbb{C}$, with isolated simple poles located at $s_k = -\alpha_j - k$ for $k \in \mathbb{N}_0$, whose residues define the discrete Barnes spectrum $\{\text{Res}_{s=s_k} \mathcal{Z}_A(s)\}_{k \in \mathbb{N}}$.

Proof. Expanding $\Phi(A)(u)$ into its Taylor series around the vertices $e_j \in \Delta_m$, the singular contributions near vertex e_j reduce to multivariable Beta integrals whose numerator contains Euler gamma functions $\Gamma(s + \alpha_j + k)$. These gamma functions possess simple poles at $s + \alpha_j + k = 0, -1, -2, \dots$ with non-vanishing residues, yielding the discrete Barnes residue spectrum. $\square$

Corollary 7.3 (OBL-019: Kigami Fractal Spectral Dimension Identity).

For a self-similar fractal realization generated by iterated simplex decimations with N cells and harmonic resistance scaling factor $\rho \in (0, 1)$, the dominant pole s_0 of the renormalized Mellin transform gives the exact Kigami spectral dimension:

$$s_0 = -\frac{d_s}{2} = -\frac{\log N}{\log(N/\rho)} \implies d_s = \frac{2 \log N}{\log(N/\rho)}. \quad (7.2)$$

Proof. Under Kigami's decimation of the resistance form on the fractal simplex [14], the heat kernel trace scales asymptotically as $t^{-d_s/2}$ where $d_s = \frac{2 \log N}{\log(N/\rho)}$. Taking the Mellin transform links the heat trace directly to the partition function poles, placing the dominant pole at $s_0 = -d_s/2$. For the standard benchmark of the Sierpinski gasket ($N = 3, \rho = 3/5$), this yields $N/\rho = 3/(3/5) = 5$, hence $d_s = \frac{2 \log 3}{\log 5} \approx 1.3652$, in exact agreement with Kigami's theorem. $\square$

8. FEDERER REACH AND MEDIAL AXIS GEOMETRY OF LEVEL SURFACES

Theorem 8.1 (OBL-020: Level Set Federer Reach Lower Bound). *Let $\Phi(A) \in C^2(\Omega, \mathbb{R})$. For any regular value t with $\epsilon_0 := \inf_{\Sigma_t} \|\nabla \Phi(A)\| > 0$ and $M := \sup_{\Sigma_t} \|\nabla^2 \Phi(A)\|_{\text{op}} < \infty$, let $d_{\text{sep}}(\Sigma_t) := \inf\{\|x - y\| \mid x, y \in \Sigma_t, \nu(x) = -\nu(y)\} > 0$ be the non-local bottleneck separation distance. Then the Federer reach of $\Sigma_t = \Phi(A)^{-1}(t)$ satisfies:*

$$\text{reach}(\Sigma_t) \geq \min\left(\frac{\epsilon_0}{M}, \frac{1}{2}d_{\text{sep}}(\Sigma_t)\right) > 0. \quad (8.1)$$

When the sublevel set $K_t(A)$ is strictly convex, $d_{\text{sep}}(\Sigma_t) = \infty$, yielding the pure curvature bound $\text{reach}(\Sigma_t) \geq \frac{\epsilon_0}{M}$.

Proof. The second fundamental form Π of the regular level set Σ_t satisfies:

$$\|\Pi(v, v)\| \leq \frac{\|\nabla^2 \Phi(A)\|_{\text{op}}}{\|\nabla \Phi(A)\|} \leq \frac{M}{\epsilon_0} \quad (8.2)$$

for any unit tangent vector $v \in T_x \Sigma_t$. By Federer's theorem [5], the reach of a compact C^2 submanifold is determined by the minimum between the local radius of curvature $\kappa_{\text{max}}^{-1} \geq \epsilon_0/M$ and half the non-local bottleneck distance $\frac{1}{2}d_{\text{sep}}(\Sigma_t)$. For strictly convex level sets, no two distinct points share anti-parallel normals with connecting segment, so $d_{\text{sep}} = \infty$, reducing the bound to ϵ_0/M . $\square$

Theorem 8.2 (OBL-021: Medial Axis Avoidance and Weyl–Federer Tube Volume Expansion). *Let Σ_t have reach $R_t > 0$ established by Theorem 8.1. For any $r < R_t$, the tubular neighborhood $U_r(\Sigma_t)$ does not intersect the medial axis $\mathcal{M}(\Phi(A))$, the metric projection $\text{Proj}_{\Sigma_t}: U_r \rightarrow \Sigma_t$ is single-valued and Lipschitz, and the tube volume satisfies the exact polynomial expansion:*

$$\text{Vol}_d(U_r(\Sigma_t)) = \sum_{k=0}^{\lfloor \frac{d-1}{2} \rfloor} \frac{2r^{2k+1}}{2k+1} \int_{\Sigma_t} H_{2k}(x) d\mathcal{H}^{d-1}(x), \quad (8.3)$$

where H_{2k} are the even elementary symmetric functions of principal curvatures.

Proof. By definition of reach, the set of points with non-unique projection $\text{Unp}(\Sigma_t)^c$ has distance at least R_t from Σ_t . Thus for $r < R_t$, $U_r \cap \mathcal{M} = \emptyset$. The volume of the tubular neighborhood is obtained by integrating the Jacobian of the normal exponential map, which by Weyl's tube formula expands into integrated curvature forms H_{2k} [5]. $\square$

9. GRAND INVARIANT TAXONOMY AND SYNTHESIS

Table 1 summarizes the evolutionary taxonomy of geometric and topological measures on functional realizations across the entire trilogy.

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TABLE 1. Grand Taxonomy of Invariants across Volumes I, II, and III.

Invariant	Codomain	Geometric Structure	Symmetry Group	Volume
$\lambda_j(A), \text{Tr}(A)$	$\mathbb{R}^n$	Euclidean / Linear	$O(n)/U(n)$	Classical
$\ \Phi(A)\ _{W^{1,p}}$	$\mathbb{R}$	Sobolev / BV	Diffeomorphisms	Vol. I
$\mathcal{W}_2(\mu_A, \mu_B)$	$\mathbb{R}_+$	Optimal Transport	Diff-preserving	Vol. II
$\kappa_{\text{BE}}(A)$	$\mathbb{R}$	Bakry–Émery Curvature	Conformal	Vol. II
$\text{Dgm}_k(A)$	$\mathbb{R}$	Persistent Homology	Homeomorphisms	Vol. II
$\mathcal{W}(\Phi(A))$	$\mathbb{R}$	Willmore Bending	Conformal	Vol. II
$c_k^{\text{EH}}(A, t), c_{\text{HZ}}$	$\mathbb{R}$	Symplectic Capacity	$\text{Symp}(\mathbb{R}^{2n}, \omega_0)$	Vol. III
$CC(\mathcal{F}_A)$	$\text{Sym}_n(\mathbb{R})$	Microlocal Sheaves	Stratified Diffeo	Vol. III
$\lambda_1^{(p)}(A)$	$\mathbb{R}$	Nonlinear Finsler	Monotone	Vol. III
$\delta(A)$	$\mathbb{R}$	Gromov Hyperbolic	Coarse Quasi-isom	Vol. III
$\mathcal{L}(A, B)$	$\mathbb{R}_+$	Non-Equilibrium Thermo	Gauge Flow	Vol. III
K_{Ω_1}, S_R, E_W	$L^2(\Omega \times \Omega)$	Holographic Entanglement	Unitary Bipartite	Vol. III
$\mathcal{Z}_A(s), s_0$	Δ_m	Barnes / Kigami Fractal	Simplicial Scale	Vol. III
$\text{reach}(\Sigma_t)$	$\mathbb{R}$	Federer Medial Axis	Isometric Tube	Vol. III

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PROGRAMA DE PÓS-GRADUAÇÃO EM ENGENHARIA DE SISTEMAS E AUTOMAÇÃO (PPGEEAA), DEPARTAMENTO DE ENGENHARIA DE SISTEMAS (DES), UNIVERSIDADE FEDERAL DE LAVRAS (UFLA), LAVRAS, MG, BRAZIL

Email address: reinaldo.silva@ufla.br